Homework #4

Math 104: Intro to Analysis

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Contents

0 Various Theorems

0.1 Limit Proof I

For $p, q, \alpha, C \in \mathbb{R}$ where $\alpha > 1$ and $C > 0$, $\lim_{q \to p} C\left|p - q\right|^{\alpha - 1} = 0$.

Proof. Let $\varepsilon > 0$. Let $\delta = \sqrt[\alpha-1]{\frac{\varepsilon}{C}}$, which is greater than zero because $\alpha - 1 > 0$ and $C > 0$. Suppose that $0 < |p - q| < \delta$. Substitute in our value of δ to find $0 < |p - q| < \sqrt[\alpha-1]{\frac{\varepsilon}{C}}$. Now, raise all sides to the positive power $\alpha - 1$, then multiply by C.

$$0 < |p - q| < \sqrt[\alpha-1]{\frac{\varepsilon}{C}} \tag{1}$$

$$0 < |p - q|^{\alpha - 1} < \frac{\varepsilon}{C} \tag{2}$$

$$0 < C\,|p - q|^{\alpha - 1} < \varepsilon \tag{3}$$

Since $0 < C\,|p - q|^{\alpha - 1}$, $C\,|p - q|^{\alpha - 1} = \left|C\,|p - q|^{\alpha - 1}\right|$. Therefore, $\left|C\,|p - q|^{\alpha - 1}\right| < \varepsilon$. Ergo, $\lim_{q \to p} C\,|p - q|^{\alpha - 1} = 0$. OE$\Delta$

0.2 Limit Proof II

$$\lim_{x \to \infty} \frac{1}{x} = 0 \tag{4}$$

Proof. Let $\varepsilon > 0$. Let $\delta = \frac{1}{\varepsilon} > 0$. Suppose that $x > \delta$. Multiply both sides by the reciprocals of x and δ to find that $\frac{1}{\delta} > \frac{1}{x}$. Substitute in the value of δ to find that $\frac{1}{\frac{1}{\varepsilon}} = \varepsilon > \frac{1}{x}$. Since $\delta > 0$ and $x > \delta$, $x > 0$, meaning $\frac{1}{x} > 0$, so $\frac{1}{x} = \left|\frac{1}{x}\right|$. So $\varepsilon > \left|\frac{1}{x}\right|$. Therefore, $x > \delta$ implies $\varepsilon > \left|\frac{1}{x}\right|$. Ergo, $\lim_{x \to \infty} \frac{1}{x} = 0$. OE$\Delta$

1 Problem 1

Let (X, d_X) and (Y, d_Y) be metric spaces. A function $f : (X, d_X) \to (Y, d_Y)$ is called *Lipschitz continuous* if there is a constant $L > 0$ so that

$$d_Y(f(p), f(q)) \leq L d_X(p, q)$$

for all $p, q \in X$. For $\alpha > 0$, we say f is *Hölder continuous (with exponent α)* if there is a constant $C > 0$ so that

$$d_Y(f(p), f(q)) \leq C d_X(p, q)^\alpha$$

for all $p, q \in X$. In particular, Hölder continuous with exponent 1 means the same thing as Lipschitz.

(a) Fix a point $p \in X$. Show that the map $g : X \to \mathbb{R}$ defined by $g(x) = d(x, p)$ is Lipschitz continuous, with constant $L = 1$.

(b) Prove that any Hölder continuous function (with any exponent $\alpha > 0$) is uniformly continuous.

(c) Prove that if f is Lipschitz continuous and bounded, then f is Hölder continuous with exponent α, for any $0 < \alpha \leq 1$.

(d) Let $[a, b]$ be a closed interval, and suppose $f : [a, b] \to \mathbb{R}$ is C^1 (meaning its first derivative exists and is continuous on $[a, b]$). Show that f is Lipschitz continuous.

(e) Suppose $f : \mathbb{R} \to \mathbb{R}$ is Hölder continuous with exponent $\alpha > 1$. Show that f is constant.

1.1 Solution (a)

Proof. Since the codomain of g is $\mathbb{R}$, we can use the standard metric as d_Y. From the triangle inequality, we know that $d_X(x, y) + d_X(p, x) \geq d_X(p, y)$ and $d_X(x, y) + d_X(p, y) \geq d_X(p, x)$. We can reorder both of these to find that $d_X(x, y) \geq d_X(p, y) - d_X(p, x)$ and $d_X(x, y) \geq d_X(p, x) - d_X(p, y)$. Consolidating these, we can apply the absolute value to find that $d_X(x, y) \geq |d_X(p, y) - d_X(p, x)|$. This is also known as the inverse triangle inequality. Since the absolute value is the usual metric over $\mathbb{R}$, this means that $d_X(x, y) \geq d_{\mathbb{R}}(d_X(p, y), d_X(p, x))$. Since $g(x) = d_X(p, x)$, $d_X(x, y) \geq d_{\mathbb{R}}(g(y), g(x))$. Therefore, g is Lipschitz continuous with constant $L = 1$. OEΔ

1.2 Solution (b)

Proof. Let f be Hölder continuous with exponent α for $\alpha > 0$.

$$d_Y(f(p), f(q)) \leq C d_X(p, q)^\alpha \tag{5}$$

Let $\varepsilon > 0$ and let $\delta = \sqrt[\alpha]{\frac{\varepsilon}{C}}$, which allows that $\delta > 0$ since $C > 0$. Suppose that for some $p, q \in X$, $d_X(p, q) < \delta$. We can substitute in our value of δ and isolate ε.

$$d_X(p, q) < \delta = \sqrt[\alpha]{\frac{\varepsilon}{C}} \tag{6}$$

$$d_X(p, q)^\alpha < \frac{\varepsilon}{C} \tag{7}$$

$$C d_X(p, q)^\alpha < \varepsilon \tag{8}$$

Since $d_Y(f(p), f(q)) \leq C d_X(p, q)^\alpha$, this means $d_Y(f(p), f(q)) < \varepsilon$. Therefore, $d_X(p, q) < \delta$ implies $d_Y(f(p), f(q)) < \varepsilon$. Ergo, f is uniformly continuous. OEΔ

1.3 Solution (c)

Proof. Let $0 < \alpha \leq 1$, and let f be Lipschitz continuous. Let $p, q \in X$.

$$d_Y(f(p), f(q)) \leq L d_X(p, q) \tag{9}$$

Since f is bounded, then there exists a number $M \in \mathbb{R}$ and a point $f(q) \in Y$ for some $q \in X$ such that for all $p \in X$, $d_Y(f(p), f(q)) < M$. Suppose we take three cases: $d_X(p,q) = 1$, $d_X(p,q) > 1$, and $d_X(p,q) < 1$.

Case 1: $d_X(p,q) = 1$**.** Anything real to the power of 1 is just 1, so if we set $C = M$, then it will always be true that $d_Y(f(p), f(q)) \leq C d_X(p,q)^\alpha$.

Case 2: $d_X(p,q) < 1$**.** If $d_X(p,q) < 1$, then for $0 < \alpha \leq 1$, $d_X(p,q) \leq d_X(p,q)^\alpha$. Therefore, if we set $C = L$, then $L d_X(p,q) \leq L d_X(p,q)^\alpha$, and therefore $d_Y(f(p), f(q)) \leq C d_X(p,q)^\alpha$.

Case 3: $d_X(p,q) > 1$**.** If $d_X(p,q) > 1$, then for all $0 < \alpha \leq 1$, $d_X(p,q)^\alpha > 1$. Therefore, if we set $C = M$, then $C d_X(p,q)^\alpha > M$, and by extension it will always be true that $d_Y(f(p), f(q)) < C d_X(p,q)^\alpha$.

Therefore, depending on the value of our inserts, we can take $C = \max\{M, L\}$ as a constant and f will be Hölder continuous. OEΔ

1.4 Solution (d)

Proof. $[a,b]$ is a subset of the real numbers, so we will be using the usual metric. If f is C^1, then its first derivative exists and is continuous on $[a,b]$. Intervals are compact in the real numbers, so $[a,b]$ is compact. Since continuous functions on compact sets are bounded, $f'([a,b])$ is bounded. There exists $M \in \mathbb{R}$ such that for all $x \in [a,b]$, $|f'(x)| < M$. Without loss of generality, suppose $p \leq q$. By the mean value theorem, there exists $c \in (p,q)$ so that $f'(c) = \frac{f(q)-f(p)}{q-p}$. Since $c \in [a,b]$, $|f'(x)| < M$, so $\left|\frac{f(q)-f(p)}{q-p}\right| < M$. By the limit laws, $\frac{|f(q)-f(p)|}{|q-p|} < M$. Multiply both sides by $|q-p|$ to find that $|f(q) - f(p)| < M|q-p|$. Since $\mathbb{R}$ uses the usual metric, this means for any $x, p \in X$, $d_{\mathbb{R}}(f(p), f(q)) \leq L d_{[a,b]}(p,q)$ for any $p, q \in [a,b]$. Therefore, f is Lipschitz continuous. OEΔ

1.5 Solution (e)

Proof. Let f be Hölder continuous with exponent $\alpha > 1$. There exists a constant C such that for all $p, q \in \mathbb{R}$:

$$d_{\mathbb{R}}(f(p), f(q)) \leq C d_{\mathbb{R}}(p,q)^\alpha \tag{10}$$

Note that $\mathbb{R}$ uses the usual metric.

$$|f(p) - f(q)| \leq C |p-q|^\alpha \tag{11}$$

Divide both sies by $|p-q|$.

$$\frac{|f(p) - f(q)|}{|p-q|} = \left|\frac{f(p) - f(q)}{p-q}\right| \leq C |p-q|^{\alpha-1} \tag{12}$$

By the definition of the absolute value, $\left|\frac{f(p)-f(q)}{p-q}\right| \geq 0$.

$$0 \leq \left|\frac{f(p) - f(q)}{p-q}\right| \leq C |p-q|^{\alpha-1} \tag{13}$$

Take the limit of all sides as q approaches p.

$$\lim_{q\to p} 0 \leq \lim_{q\to p} \left| \frac{f(p)-f(q)}{p-q} \right| \leq \lim_{q\to p} C\,|p-q|^{\alpha-1} \tag{14}$$

We proved in *0.1: Limit Proof I* that $\lim_{q\to p} C\,|p-q|^{\alpha-1} = 0$. Since 0 is constant and contains neither p not q, $\lim_{q\to p} 0 = 0$. Therefore, by the squeeze theorem, $\lim_{q\to p} \left| \frac{f(p)-f(q)}{p-q} \right| = f(p)' = 0$ for all $p \in \mathbb{R}$.

Now, recall the mean value theorem: if f is a real, continuous function on $[a, b]$ which is differentiable on (a, b), there is a point $x \in (a, b)$ where $f(b) - f(a) = (b-a)f'(x)$. Since our $f'(x) = 0$ for all $x \in \mathbb{R}$, fix p and let $q \in \mathbb{R}$. The Mean Value Theorem tells us that for the interval $[p, q]$, there exists a point $x \in \mathbb{R}$ such that $f(q) - f(p) = (q-p)f'(x) = (q-p) * 0 = 0$. Therefore, for any $q \in \mathbb{R}$, $f(p) = f(q)$. Ergo, f is constant. OEΔ

2 Problem 2

On the interval $(100, \infty)$, define functions $f(x) = x + \sin(x)$, $g(x) = x$, $p(x) = x + \sin(x)\cos(x)$, and $q(x) = p(x)e^{\sin(x)}$.

(a) Using facts we proved about limits and derivatives, prove that

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = 1, \quad \lim_{x\to\infty} \frac{f'(x)}{g'(x)} \text{ DNE}, \quad \lim_{x\to\infty} \frac{p(x)}{q(x)} \text{ DNE, and } \lim_{x\to\infty} \frac{p'(x)}{q'(x)} = 0.$$

(b) For both scenarios $\frac{f(x)}{g(x)}$ and $\frac{p(x)}{q(x)}$, explain why these results do not contradict L'Hôpital's rule.

Note: even though we haven't learned about sin *and* cos *yet, you are free to use any familiar properties of them you wish.*

2.1 Solution (a)

We prove that $\lim_{x\to\infty} \frac{f(x)}{g(x)} = 1$.

Proof. First write out the limit.

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = \lim_{x\to\infty} \frac{x + \sin(x)}{x} = \lim_{x\to\infty} \left[1 + \frac{\sin(x)}{x}\right] \tag{15}$$

By limit laws, we can separated added up properties.

$$\lim_{x\to\infty} \left[1 + \frac{\sin(x)}{x}\right] = \lim_{x\to\infty} 1 + \lim_{x\to\infty} \frac{\sin(x)}{x} \tag{16}$$

Since 1 is constant, $\lim_{x\to\infty} 1 = 1$. My first assertion about sin: it is bounded to $[-1, 1]$ and is cyclical, so $-1 \leq \sin(x) \leq 1$. Divide all sides by x to find $-\frac{1}{x} \leq \frac{\sin(x)}{x} \leq \frac{1}{x}$. Take the limit as x approaches ∞ to get $-\lim_{x\to\infty} \frac{1}{x} = \lim_{x\to\infty} -\frac{1}{x} \leq \lim_{x\to\infty} \frac{\sin(x)}{x} \leq \lim_{x\to\infty} \frac{1}{x}$. We proved earlier that $\lim_{x\to\infty} \frac{1}{x} = 0$, and by extension, $-\lim_{x\to\infty} \frac{1}{x} = 0$. Therefore, by the sequence squeeze theorem, $\lim_{x\to\infty} \frac{\sin(x)}{x} = 0$. Put this back into $\lim_{x\to\infty} \frac{f(x)}{g(x)}$.

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = \lim_{x\to\infty} 1 + \lim_{x\to\infty} \frac{\sin(x)}{x} = 1 + 0 = 1 \tag{17}$$

OEΔ

We prove that $\lim_{x\to\infty} \frac{f'(x)}{g'(x)}$ does not exist.

Proof. My second assertion is that if $f(x) = \sin(x)$, $f'(x) = \cos(x)$. Since $(a + b)' = a' + b'$, $(x + \sin(x))' = (x') + (\sin(x))' = 1 + \cos(x)$.

$$\lim_{x\to\infty} \frac{f'(x)}{g'(x)} = \lim_{x\to\infty} [1 + \cos(x)] \tag{18}$$

Create two sequences in terms of $n \in \mathbb{N}$ that diverge to ∞ as $n \to \infty$.

$$a_n = 2n\pi; b_n = (2n + 1)\pi \tag{19}$$

Apply both a_n and b_n to $1+\cos(x)$.

$$1+\cos(a_n) = 1+\cos(2n\pi) = 1+1 = 2 \tag{20}$$
$$1+\cos(b_n) = 1+\cos((2n+1)\pi) = 1-1 = 0 \tag{21}$$

This gives us two subsequences that converge to two different values as $x \to \infty$. Therefore, the limit does not exist. OEΔ

We prove that $\lim_{x\to\infty} \frac{p(x)}{q(x)}$ does not exist.

Proof.

$$\lim_{x\to\infty} \frac{p(x)}{q(x)} = \lim_{x\to\infty} \frac{p(x)}{p(x)e^{\sin(x)}} = \lim_{x\to\infty} e^{-\sin(x)} \tag{22}$$

Define two sequences as $n \to \infty$.

$$a_n = \left(2n+\frac{1}{2}\right)\pi; b_n = \left(2n+\frac{3}{2}\right)\pi$$

Apply both a_n and b_n to $e^{-\sin(x)}$.

$$e^{-\sin(a_n)} = e^{-\sin\left(\left(2n+\frac{1}{2}\right)\pi\right)} = e^{-1} \tag{23}$$
$$e^{-\sin(b_n)} = e^{-\sin\left(\left(2n+\frac{3}{2}\right)\pi\right)} = e^{1} = e \tag{24}$$

This gives us two subsequences that converge to two different values as $x \to \infty$. Therefore, the limit does not exist. OEΔ

We prove that $\lim_{x\to\infty} \frac{p'(x)}{q'(x)} = 0$.

Proof. First, we find the derivative of $p(x)$ and $q(x)$. This involves the multiplication rule ($(fg)' = fg' + f'g$), the chain rule ($(f(g))' = g'f'(g)$), my assertion that $(\cos(x))' = -\sin(x)$, and the . I was given verbal approval in Office Hours to assert that $\frac{\mathrm{d}}{\mathrm{d}x}e^x = e^x$.

$$\begin{aligned} p'(x) &= (x+\sin(x)\cos(x))' = 1 + (\sin(x)\cos(x))' = 1+\cos^2(x)-\sin^2(x) && (25)\\ &= 1+\cos(2x) = 2\cos^2(x) && (26)\\ q'(x) &= \left(p(x)e^{\sin(x)}\right)' = p'(x)e^{\sin(x)} + p(x)\left(e^{\sin(x)}\right)' && (27)\\ &= p'(x)e^{\sin(x)} + p(x)\cos(x)e^{\sin(x)} && (28)\\ &= 2\cos^2(x)e^{\sin(x)} + (x+\sin(x)\cos(x))\cos(x)e^{\sin(x)} && (29)\\ &= e^{\sin(x)}\cos(x)\left(2\cos(x)+x+\sin(x)\cos(x)\right) && (30)\end{aligned}$$

Now, we compute $\frac{p'(x)}{q'(x)}$.

$$\begin{aligned} \frac{p'(x)}{q'(x)} &= \frac{2\cos^2(x)}{e^{\sin(x)}\cos(x)\left(2\cos(x)+x+\sin(x)\cos(x)\right)} && (31)\\ &= \frac{2\cos(x)}{e^{\sin(x)}\left(2\cos(x)+x+\sin(x)\cos(x)\right)} && (32)\end{aligned}$$

Take a few inequalities based on the bounds of $\sin(x)$ and $\cos(x)$ being $-1 \leq \sin(x) \leq 1$ and $-1 \leq \cos(x) \leq 1$.

$$e^{-1} \leq e^{\sin(x)} \leq e \tag{33}$$

$$-2 \leq 2\cos(x) \leq 2 \tag{34}$$

$$-1 \leq \sin(x)\cos(x) \leq 1 \tag{35}$$

$$-3 \leq 2\cos(x) + \sin(x)\cos(x) \leq 3 \tag{36}$$

$$x - 3 \leq 2\cos(x) + x + \sin(x)\cos(x) \leq x + 3 \tag{37}$$

$$e^{-1}\left(x-3\right) \leq e^{\sin(x)}\left(2\cos(x) + x + \sin(x)\cos(x)\right) \leq e\left(x+3\right) \tag{38}$$

$$\frac{1}{e^{-1}\left(x-3\right)} \geq \frac{1}{e^{\sin(x)}\left(2\cos(x) + x + \sin(x)\cos(x)\right)} \geq \frac{1}{e\left(x+3\right)} \tag{39}$$

OEΔ

3 Problem 3

Let $f : (a, b) \to \mathbb{R}$ be continuous. Let $c \in (a, b)$, and suppose f is differentiable on (a, c) and (c, b). Suppose that

$$\lim_{x \to c} f'(x)$$

exists and is a real number. Show that f is differentiable at c. *Hint:* Mean value theorem.

3.1 Solution

Proof. Let $\lim_{x \to c} f'(x) = L$. Recall the definition of $f'(c)$.

$$f'(c) = \lim_{t \to c} \frac{f(t) - f(c)}{t - c} \tag{40}$$

Without loss of generality, suppose that $t > c$. By the Mean Value Theorem, there exists $d \in (c, t)$ where $\frac{f(t)-f(c)}{t-c} = f'(d)$. Since $d \in (c, t)$, $c < d < t$, so $0 < d - c < t - c$ and therefore $0 < |d - c| < |t - c|$.

$f'(c) = \lim_{t \to c} \frac{f(t)-f(c)}{t-c}$ means that for all $\varepsilon > 0$, there exists $\delta > 0$ where $|t - c| < \delta$ implies $\left| \frac{f(x)-f(c)}{x-c} - f'(c) \right| < \varepsilon$. Since $|d - c| < |t - c|$, $|d - c| < \delta$. Furthermore, since $\frac{f(t)-f(c)}{t-c} = f'(d)$, $|f'(d) - f'(c)| < \varepsilon$. Therefore, for all $\varepsilon > 0$, there exists $\delta > 0$ where $|d - c| < \delta$ implies $|f'(d) - f'(c)| < \varepsilon$. This means $f'(c) = \lim_{d \to c} f'(d)$. It is given that $\lim_{x \to c} f'(x)$ exists and is a real number. Therefore $f'(c)$ exists and is a real number. Ergo, f is differentiable at c. OEΔ

4 Problem 4

Let $f : \mathbb{R} \to \mathbb{R}$ be C^2 (meaning its second derivative exists and is continuous on $\mathbb{R}$).

(a) Let $x \in \mathbb{R}$. For each $h > 0$, show there are numbers ξ and η with $x - h < \xi < x < \eta < x + h$ and

$$\frac{f''(\xi) + f''(\eta)}{2} = \frac{f(x+h) + f(x-h) - 2f(x)}{h^2}.$$

(b) Show that, for each $x \in \mathbb{R}$,

$$f''(x) = \lim_{h \to 0} \frac{f(x+h) + f(x-h) - 2f(x)}{h^2}.$$

(c) A function $f : \mathbb{R} \to \mathbb{R}$ is called *convex* if for all $x, y \in \mathbb{R}$ and $0 \leq \lambda \leq 1$, we have

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y).$$

Geometrically, this means the graph of f lies *below* all of its secant lines. Show that any C^2 convex function f satisfies $f''(x) \geq 0$ for all x. *Hint:* convexity tells you something about the numerator in part (b).

(d) Conversely, if a twice-differentiable function f satisfies $f''(x) \geq 0$ for all x, show that f is convex. *Hint:* For $\lambda \in (0,1)$ and $x < y \in \mathbb{R}$, the number $z := \lambda x + (1-\lambda)y$ satisfies $x < z < y$. Apply the mean value theorem on $[x, z]$ and $[z, y]$ to get an expression for $\lambda f(x) + (1-\lambda)f(y)$.

5 Problem 5

Let $f : [0,1] \to \mathbb{R}$ be defined by

$$f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q} \\ 0, & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Compute the upper and lower Riemann integrals of f. Is f Riemann integrable?

5.1 Solution

The issue here is the density of the rationals in the reals.

First we prove the upper Riemann integral.

$$\overline{\int_0^1} f(x)dx = 1 \tag{41}$$

Proof. Let P be any partition of $[0,1]$ with $n+1$ elements. Take any $j \in \mathbb{N}$ where $1 \le j \le n$. By the density of the rationals in the reals, there exists $x \in [x_{j-1}, x_j]$ where $x \in \mathbb{Q}$. Since the only two possible values of $f(x)$ are 0 and 1, $\sup_{x \in [x_{j-1}, x_j]} f(x) = 1$. We can compute the upper sum of f from here.

$$U(P,f) = \sum_{j=1}^{n} \Delta x_j \sup_{t \in [x_{j-1}, x_j]} f(t) = \sum_{j=1}^{n} \Delta x_j * 1 = \sum_{j=1}^{n} \Delta x_j \tag{42}$$

Since the sum of Δx_j telescopes, $\sum_{j=1}^{n} \Delta x_j = x_n - x_0$. Therefore, for any partition P of $[0,1]$, $U(P,f) = 1 - 0 = 1$. Since $U(P,f) = 1$ for every partition P of $[0,1]$, $\inf\{U(P,f) | \text{Partition } P\} = 1$. Ergo, $\overline{\int_0^1} f(x)dx = \inf\{U(P,f) | \text{Partition } P\} = 1$. OEΔ

Now we prove the lower Riemann integral.

$$\underline{\int_0^1} f(x)dx = 0 \tag{43}$$

Proof. Let P be any partition of $[0,1]$ with $n+1$ elements. Take any $j \in \mathbb{N}$ where $1 \le j \le n$. By the density of the irrationals in the reals, there exists $t \in [x_{j-1}, x_j]$ where $t \notin \mathbb{Q}$. Since the only two possible values of $f(x)$ are 0 and 1, $\inf_{t \in [x_{j-1}, x_j]} f(t) = 0$. We can compute the upper sum of f from here.

$$L(P,f) = \sum_{j=1}^{n} \Delta x_j \inf_{t \in [x_{j-1}, x_j]} f(t) = \sum_{j=1}^{n} \Delta x_j * 0 = \sum_{j=1}^{n} 0 = 0 \tag{44}$$

Therefore, for any partition P of $[0,1]$, $L(P,f) = 0$. Since $L(P,f) = 0$ for every partition P of $[0,1]$, $\sup\{U(P,f) | \text{Partition } P\} = 0$. Ergo, $\underline{\int_0^1} f(x)dx = \sup\{L(P,f) | \text{Partition } P\} = 0$. OEΔ

Since $\overline{\int_0^1} f(x)dx \ne \underline{\int_0^1} f(x)dx$, the function is not Riemann integrable on $[0,1]$.